SPARTA GLOBAL | AI & TECH INSTITUTE

AI FUNDAMENTALS

DEMYSTIFYING AI: HOW IT WORKS AND WORKS FOR YOU

Explore how AI systems are built and how they operate, from neural networks and machine learning models to the practical applications and opportunities

TODAY

DEMYSTIFYING AI

Explore how AI systems are built and how they operate, from neural networks and machine learning models to the practical applications and opportunities

NEXT SESSION

UNDERSTANDING MODERN AI

Investigate the defining models used in modern AI, from the architectures that power them to the strengths, limitations, and reliability challenges of AI systems

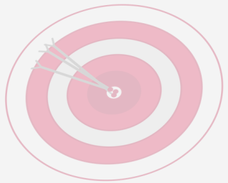
SESSION OVERVIEW

01 **Intro to AI**

02 **The Brain of AI**

03 **Classification AI Walkthrough**

LEARNING OBJECTIVES



→ Confidently define AI and its subfields

→ Understand how AI is powered, and which parameters and techniques are crucial to its success

01

INTRO TO AI

Learning the Terms, Learning the Uses

INTRO TO AI

WHAT IS IT?

- Artificial Intelligence
  - A Simulation of Human Intelligence
- Computers are taught to recognise patterns that humans can
- AI uses these learned patterns to:
  - Identify
  - Analyse
  - Manipulate
  - Create

INTRO TO AI

WHERE IS IT?

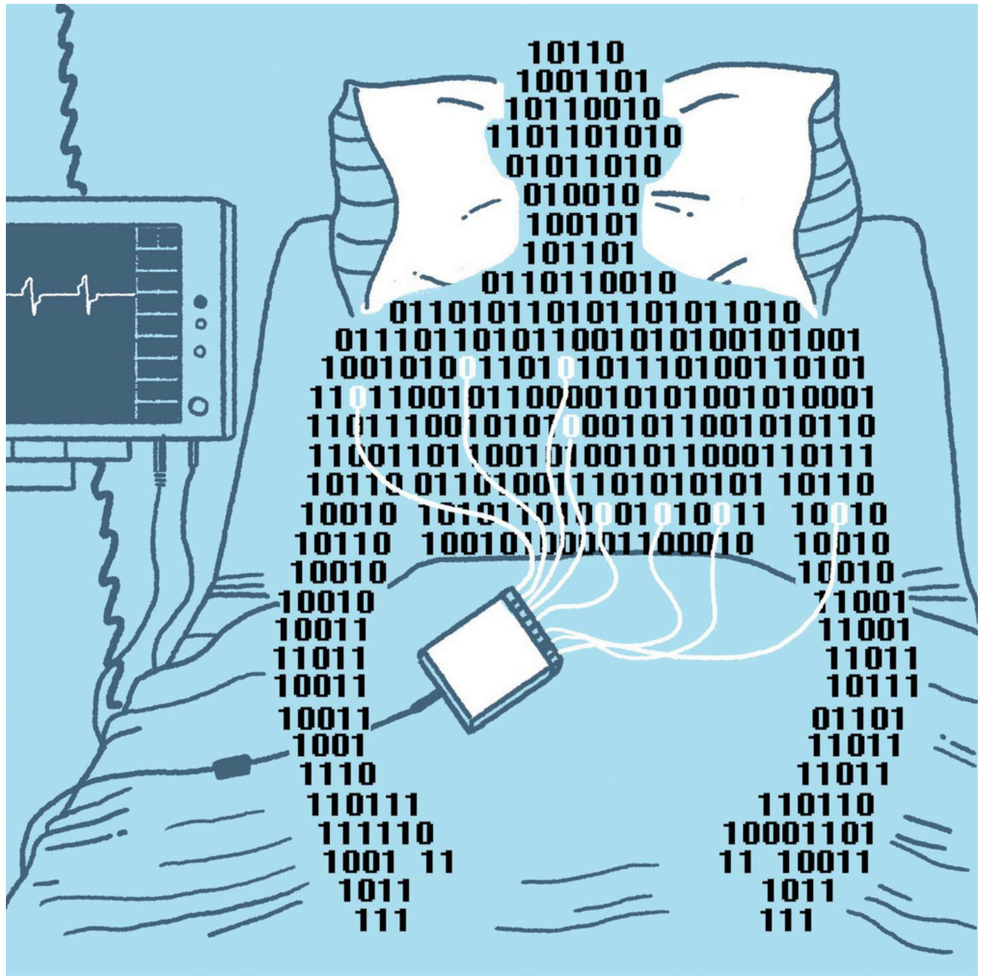
QUESTION

When's the last time you encountered AI?

- AI is now present in every realm of life, including:
  - The chatbots and assistants people use everyday - like *ChatGPT*, *Gemini* and *Siri*
  - Movie and TV show recommendations in streaming apps
  - Browsers and workplace platforms to summarise information
  - Industries such as technology, data analytics, and healthcare

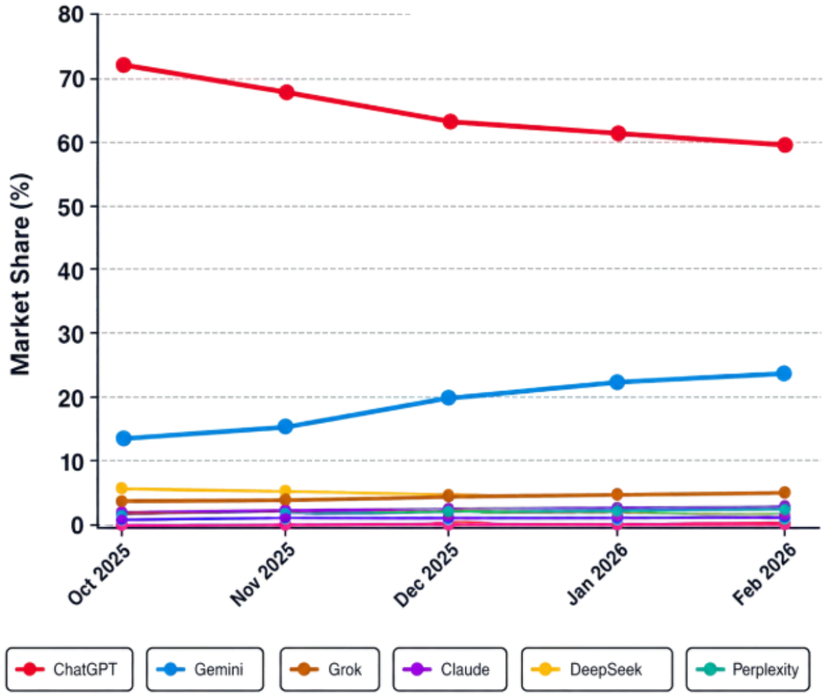
INTRO TO AI
IN HEALTHCARE

- AI is a life-changing tool in the healthcare industry
- Due to its ability to recognise patterns, it can identify and classify many medical conditions quickly and accurately
- For example, John Hopkins University's Targeted Real-Time Early Warning System for Sepsis:
  - Their solution successfuly caught symptoms within hours of patients arriving at the hospital
  - Had a significant impact, reducing patient mortality rate by 20%

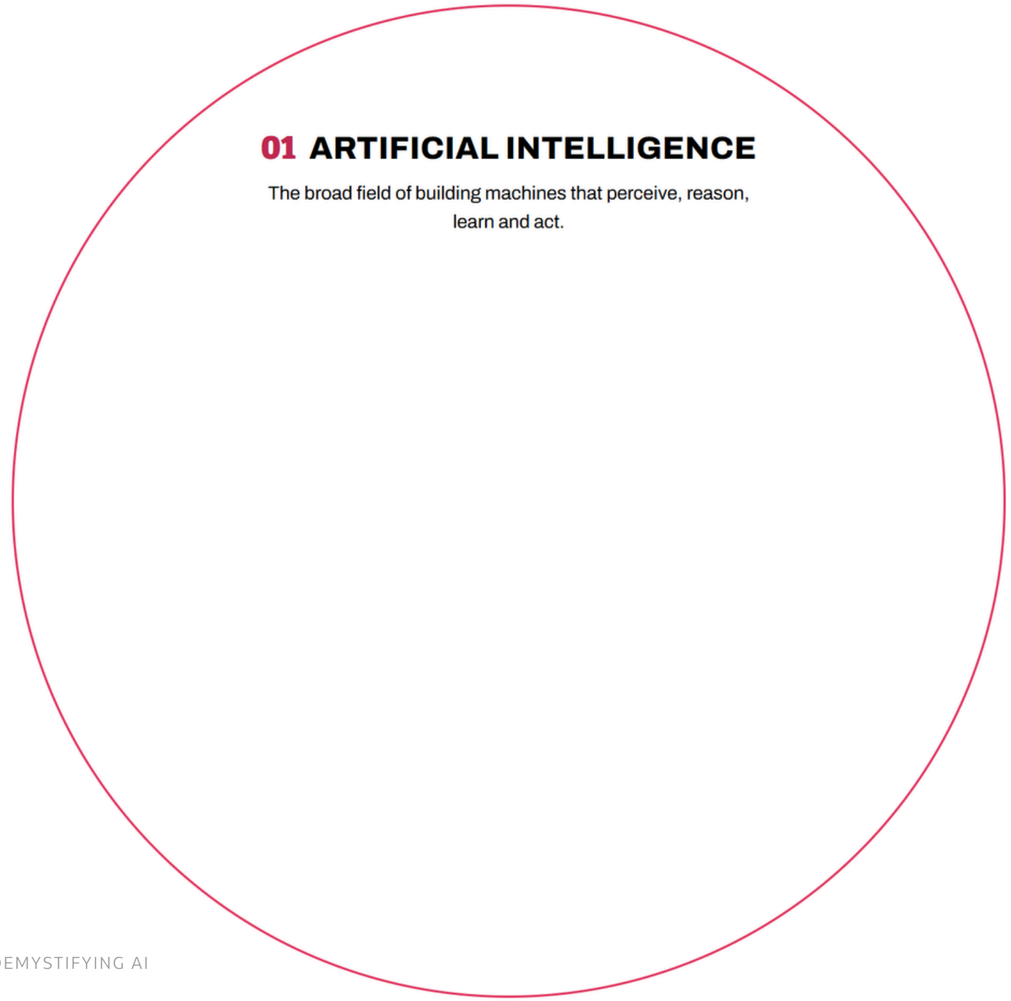


INTRO TO AI
IN CHATBOTS

- ChatGPT is by far the most-adopted AI chatbot and assistant *(Ermut, 2026)*
- As of February 2026, 60% of users choose it over others available
- This majority has been slowly decreasing over time with the rise of competition from alternatives such as:
  - Gemini, 24%
  - Grok, 3.4%
  - Claude, 3.3%
  - DeepSeek, 2.8%



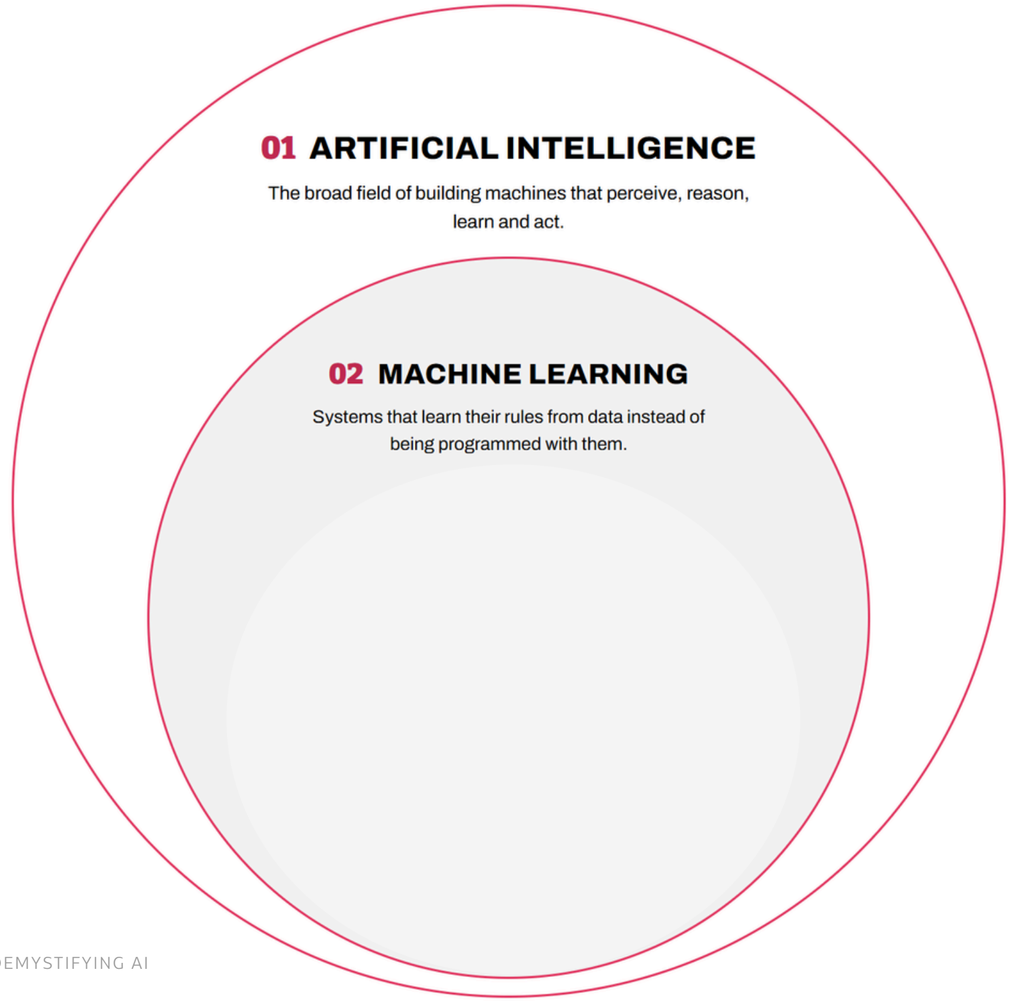
INTRO TO AI
TERMINOLOGY



01 ARTIFICIAL INTELLIGENCE

The umbrella term for technologies that let machines perceive, reason, learn and act. Rule-based expert systems count too - AI is much older than the current wave.

INTRO TO AI
TERMINOLOGY



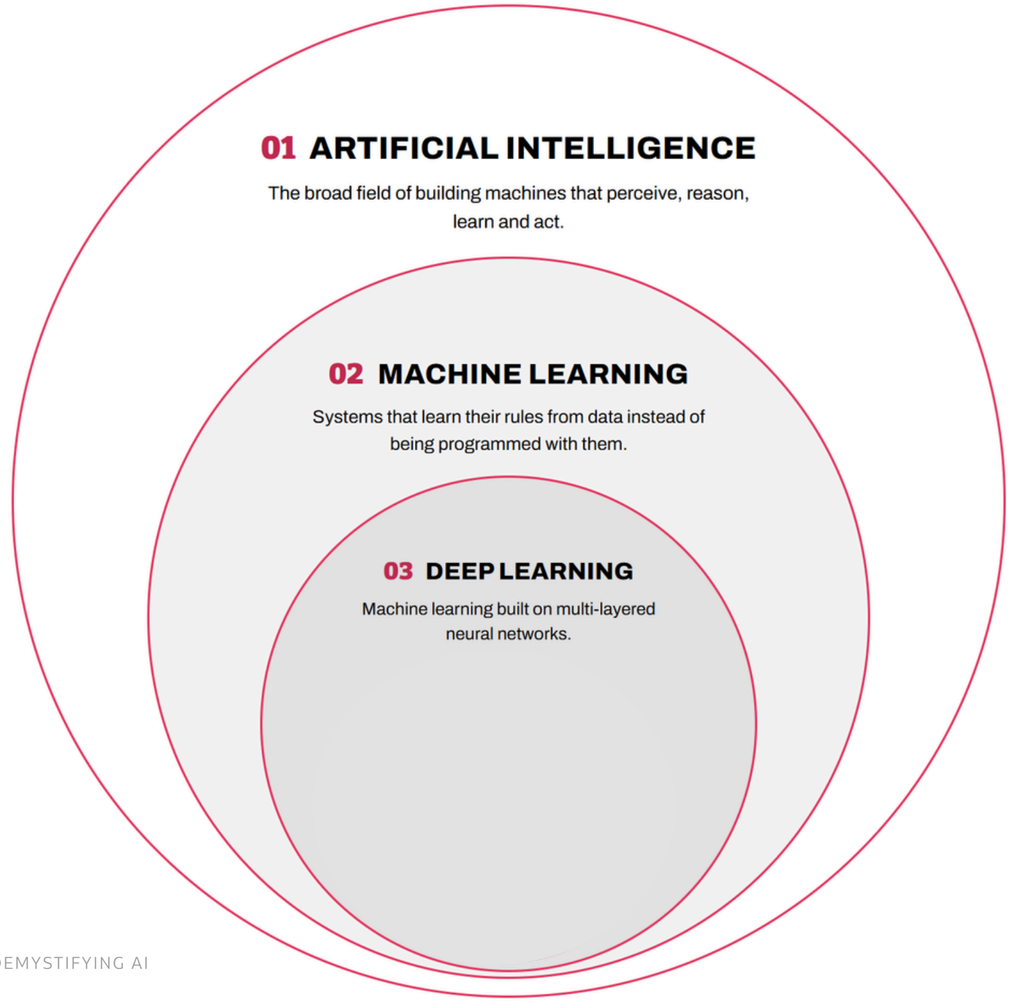
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A subset of AI in which a model learns patterns from data rather than following rules written by hand. Fraud scoring, demand forecasting and recommendation live here.

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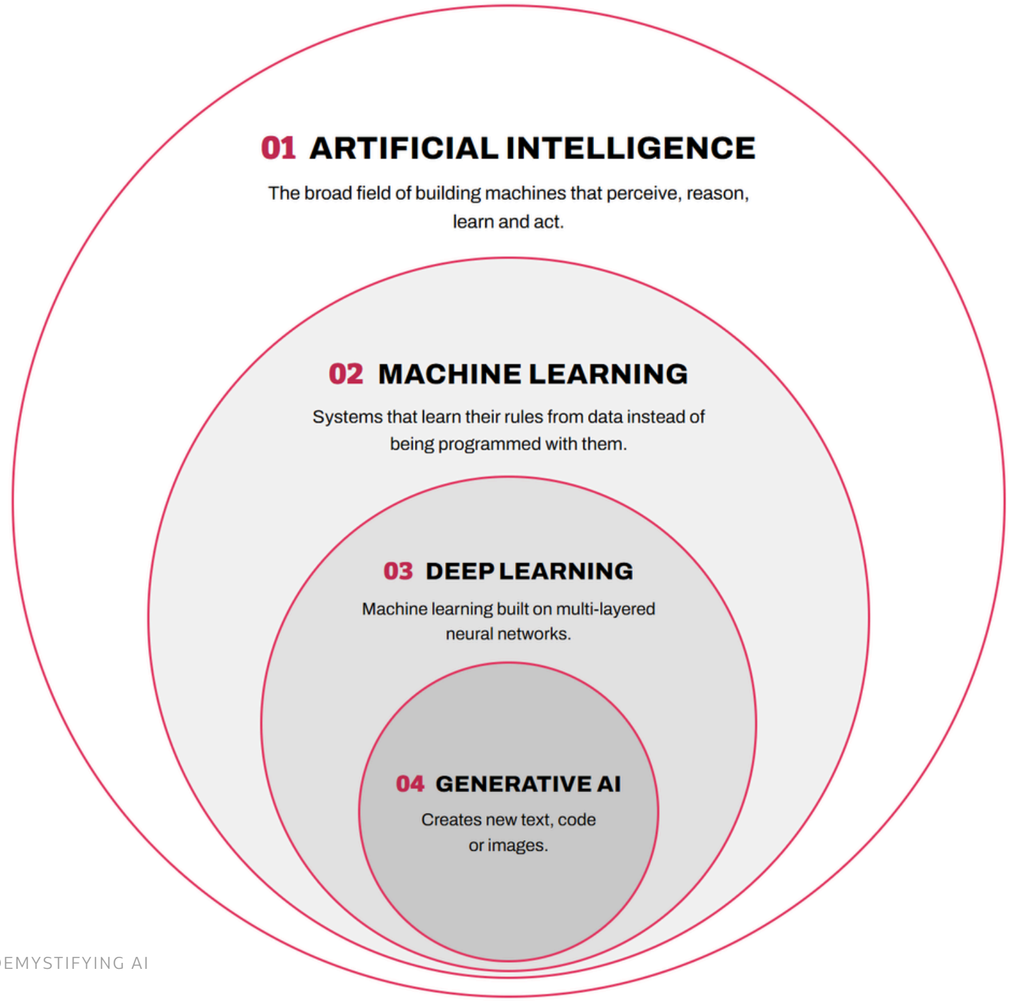
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04 GENERATIVE AI

Deep learning applied to producing something new - text, code, images from the patterns it has learned. This is the layer Spartans work with day to day.

INTRO TO AI
TERMINOLOGY

- Some specialist terms you may across in the space, are **Traditional AI** and **Modern AI**

- **Traditional AI** is AI designed and trained for a specific task
  - This is what the first iterations of AI looked like
  - Some AI (e.g. in-house AI tools) still use this model
- **Modern AI** is more generalist and highly adaptable
  - Can be adapted to new tasks without being retrained from scratch

QUESTION

Can you identify examples of both?

02

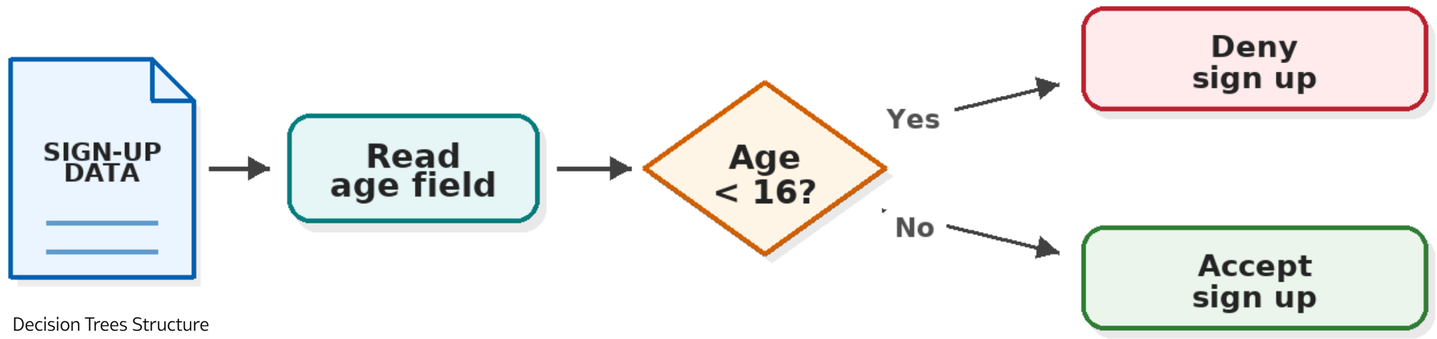
THE BRAIN OF AI

How AI Works: From Data to Decisions

THE BRAIN OF AI

BEHIND THE POWER

- AI is the overall goal of making computers as intelligent as humans
- This is supported through the AI subfield of Machine Learning (ML)
- Machine learning is how **data** is used to predict correct behaviour
- There are three main approaches ML uses to generate responses:
  - Decision Trees
    - *Learn rules*
  - Neural Networks
    - *Learn patterns*
  - Statistical Models
    - *Learn relationships*



Decision Trees Structure

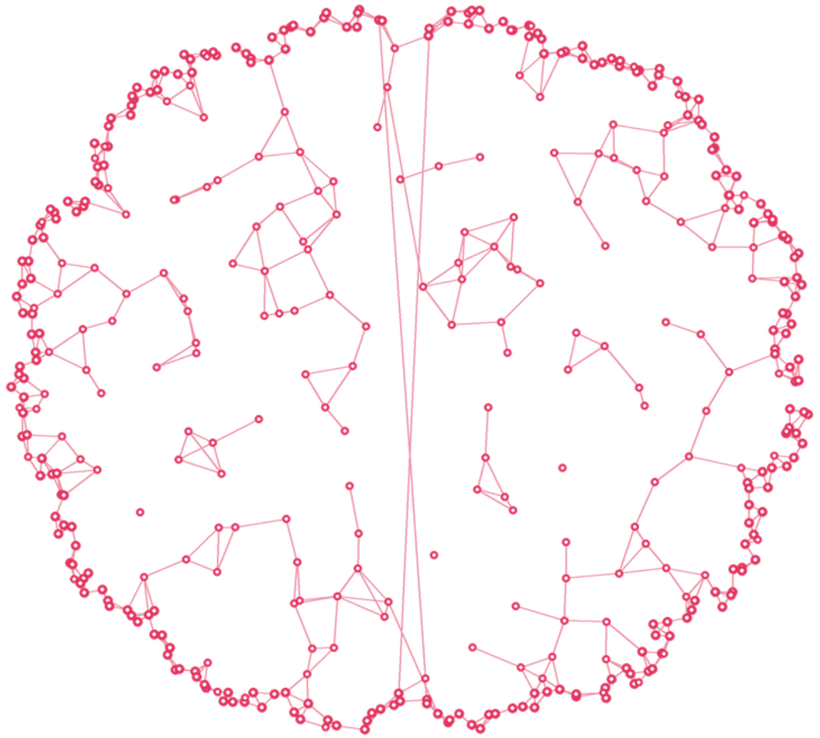
THE BRAIN OF AI

BEHIND THE POWER

- The system most AIs are built on are **neural networks**

- Named after what they aspire to be, a simulation of the neural networks in our brain
  - Today, the term is now first associated with AI rather than neuroscience

- But what are neural networks?
  - *A system composed of interlinked nodes*
  - *Nodes are arranged into layers that process information*
  - *Network learns by adjusting the strength of connections between nodes*

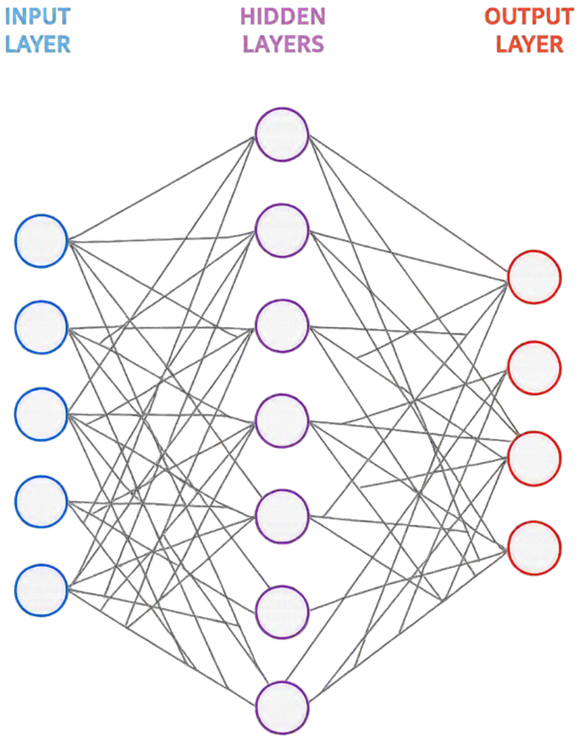


THE BRAIN OF AI

NEURAL NETWORKS

- Nodes can be arranged in different ways to specialise for different types of data processing

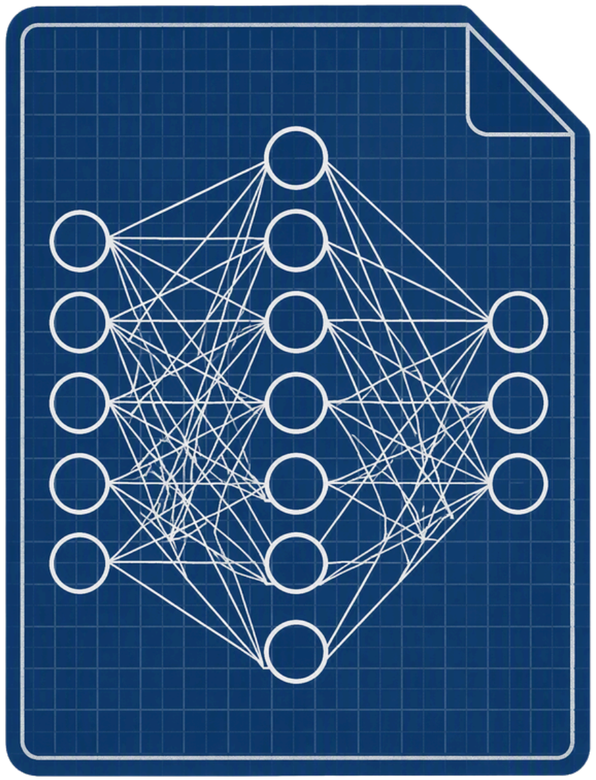
- The most common ones you may encounter are:
  - CNNs (Convolutional Neural Networks)
    - *Best for Image Classification and Computer Vision*
  - FNNs (Feed-Forward Neural Networks)
    - *Best for Structured Data Processing without Memory*
  - RNNs (Recurrent Neural Networks)
    - *Best for Sequential Data Processing with Memory*



NN architecture

THE BRAIN OF AI

MODELS



- Once you've selected a neural network architecture, it is trained on data to create a **model**
  - Think of the architecture as the blueprint and the model as the finished product
- A model is a trained neural network that is ready to perform a task
  - Examples include GPT-5.5 and Claude Opus 5
- Most commercial models are closed, meaning they cannot be modified by users
- Some are fully-open or partially-open, allowing developers to customise or further train them

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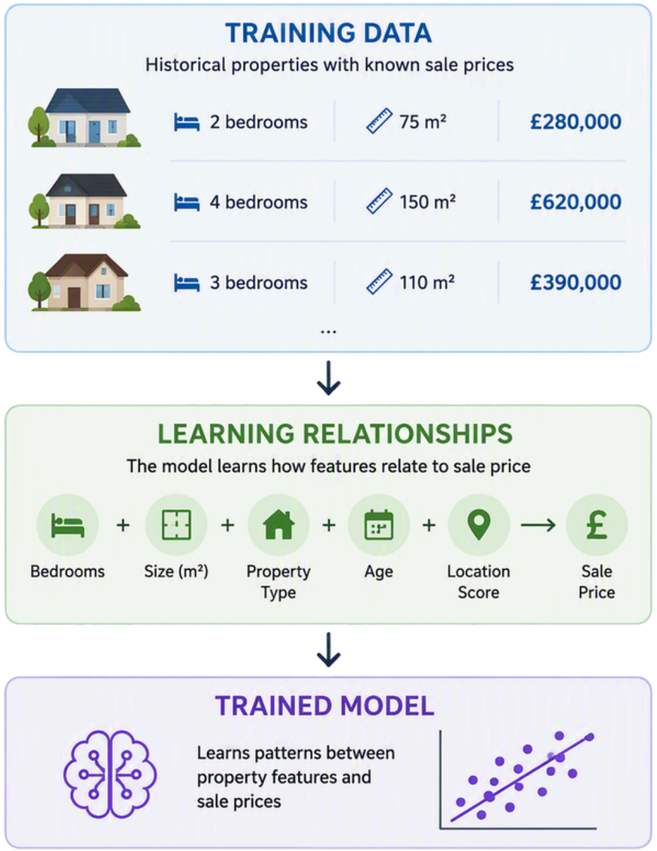
BRAIN TRAINING

- Model training is often referred to as “Learning” in the context of ML
- It describes the process of how the model will ***learn*** to approach a task in the desired way
- There are three main ways of training your model:
    - Supervised Learning
    - Unsupervised Learning
    - Reinforcement Learning

THE BRAIN OF AI

BRAIN TRAINING | SUPERVISED LEARNING

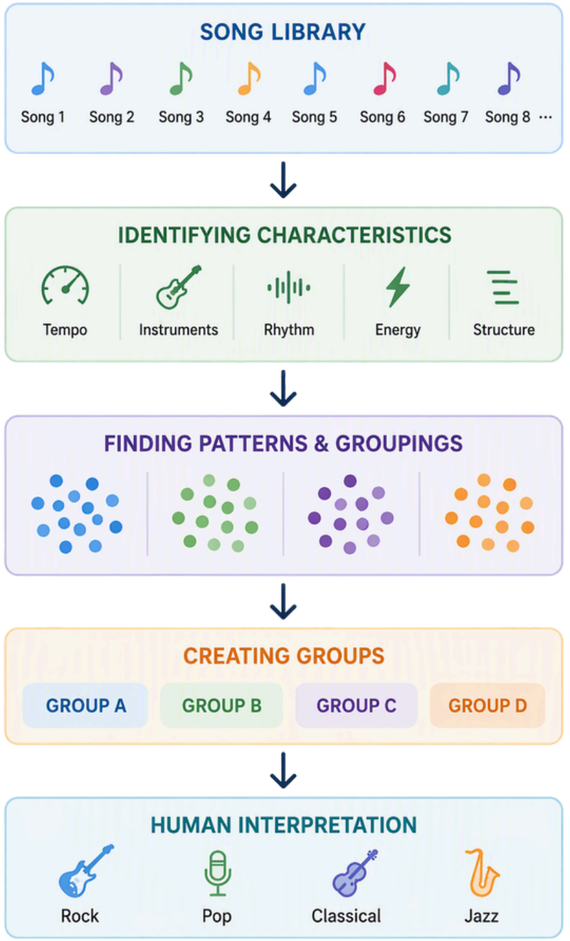
- Supervised learning trains a model using data that includes known answers or labels
- The model learns by comparing its predictions to the corrected outputs and adjusting accordingly
- This allows it to learn patterns that can later be applied to new data
- For example:
  - A model is trained on thousands of houses and their sale prices.
  - It learns the relationship between a property's characteristics and its value
  - The model can then infer the price of a house it has never seen



THE BRAIN OF AI

BRAIN TRAINING | UNSUPERVISED LEARNING

- Unsupervised learning is where a model learns from data without being given the correct answers
- It identifies patterns, relations, and groups within the data for itself
- For example:
  - A music streaming service has added a compulsory genre field to its song uploads, but all previous songs do not have this
  - To align those songs to the new system, unsupervised learning is used to identify similar characteristics to group them together
  - Humans can review *after* the training to assign labels to those classifications if needed



THE BRAIN OF AI

BRAIN TRAINING | REINFORCEMENT LEARNING

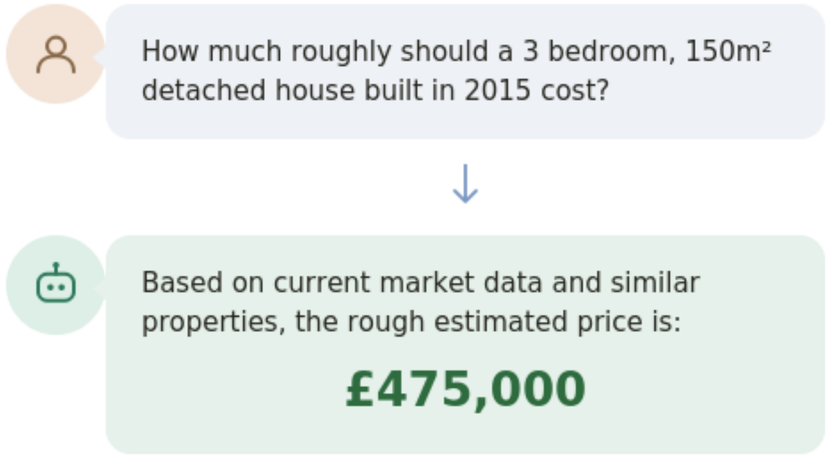
- Reinforcement learning is where a model is incentivised to “think correctly” by influence of rewards (or penalties)
- This is the technique used to train pets:
  - If you say “sit” and they sit, they get a treat
  - If you say “sit” and they do not sit, they do not get a treat
  - Eventually they realise there’s a correlation
  - As a result, they are more likely to sit when given the command
- In terms of training AI models, rewards and penalties are given as numerical values, + good result, - bad result



THE BRAIN OF AI

USING THE MODEL

- Once your model has completed training, it is ready to be used
- It utilises the model's trained weights from its learning data to infer the expected result using new data
- This stage is called *inference*
- The outcome of this stage is the final AI response



THE BRAIN OF AI

WHEN TO RETHINK A MODEL

QUESTION

When do you think a model needs to be updated or changed?

- When responses are:
  - Inaccurate
    - *e.g. putting a 1-bed house of 20m^2 as worth £1,000,000*
  - Out-of-Date
    - *e.g. not amending house prices based on the current year inflation rates*

- To remedy you would either replace the model with a newer one, e.g. GPT-5.6 replacing 5.5, or update the model through retraining (done in-house) with new or increased training data

03

CLASSIFICATION AI: A WALKTHROUGH

NN in Action: Building, Training and Using a Model

CLASSIFICATION AI

THE PURPOSE

- We want to make a classifier that can accurately tell a cat from a dog

QUESTION

What type of neural network is best suited to do this?

- We will be using a Convolutional Neural Network for this task
  - Best for images

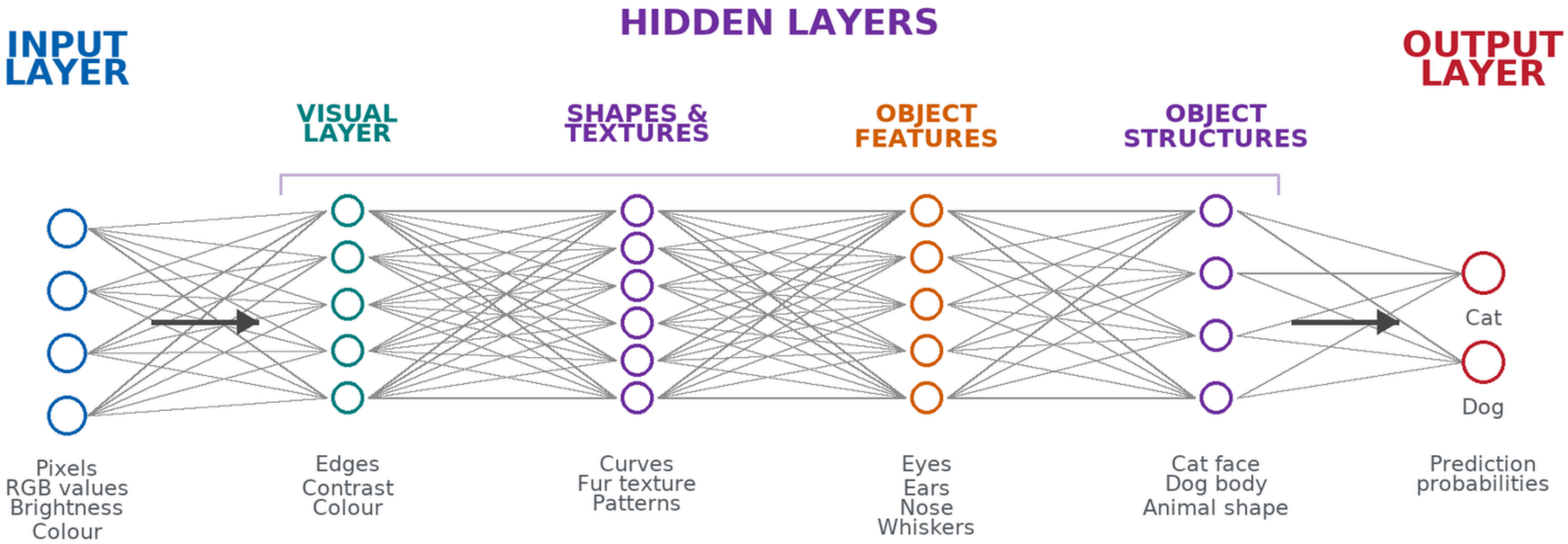
- Tech Stack:
  - Python
  - Tensorflow with Keras (A package for building ML models)



CLASSIFICATION AI

LOOKING INTO THE LAYERS

- For the CNN we will be building, we will need multiple layers to process image data
- We will be using the following architecture:



CLASSIFICATION AI

THE DATASET

- We will train the CNN using labelled data
- To do this, it's as simple as making folders with names
- The names act as the labels, and we put the relevant images in the respective folders



CLASSIFICATION AI

HOW TRAINING WORKS

Model setup

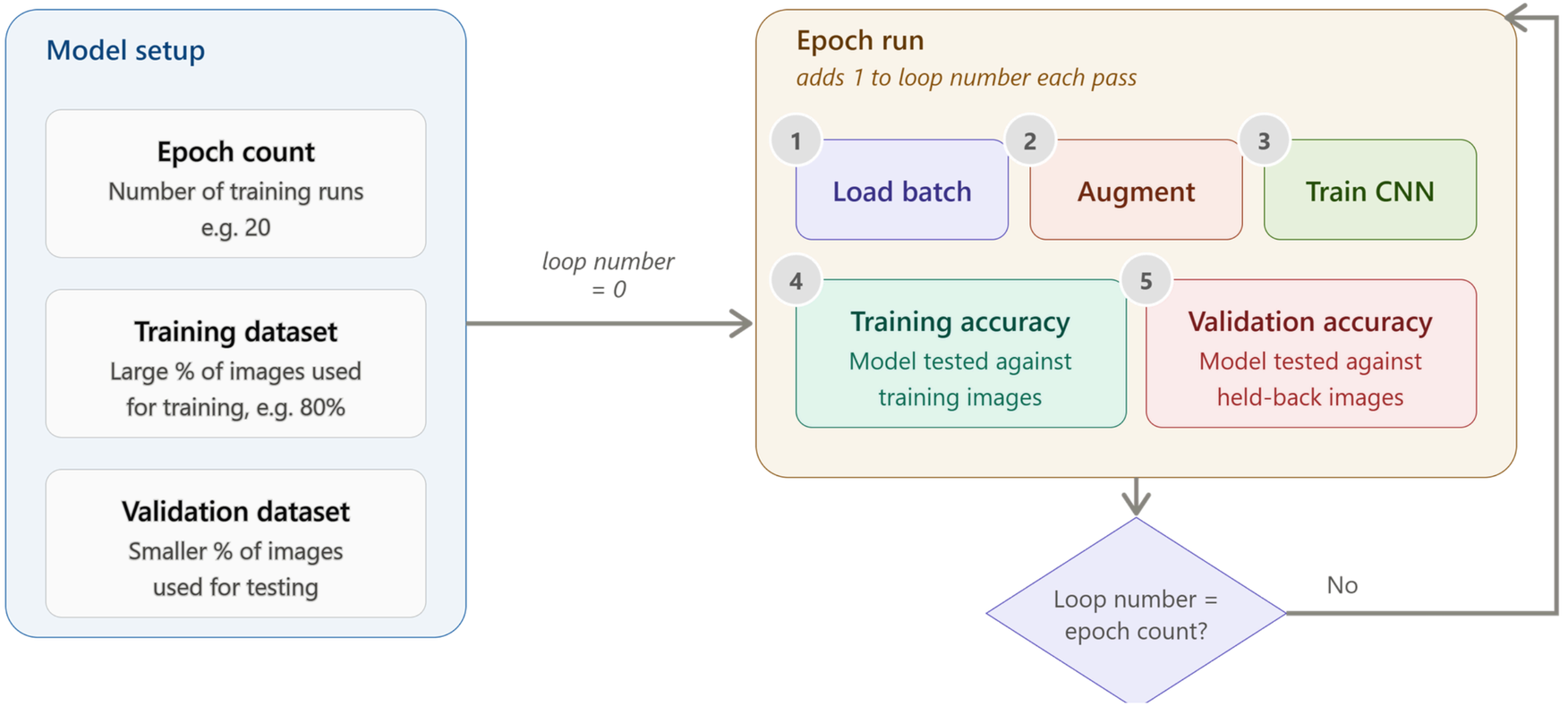
Epoch count
Number of training runs
e.g. 20

Training dataset
Large % of images used
for training, e.g. 80%

Validation dataset
Smaller % of images
used for testing

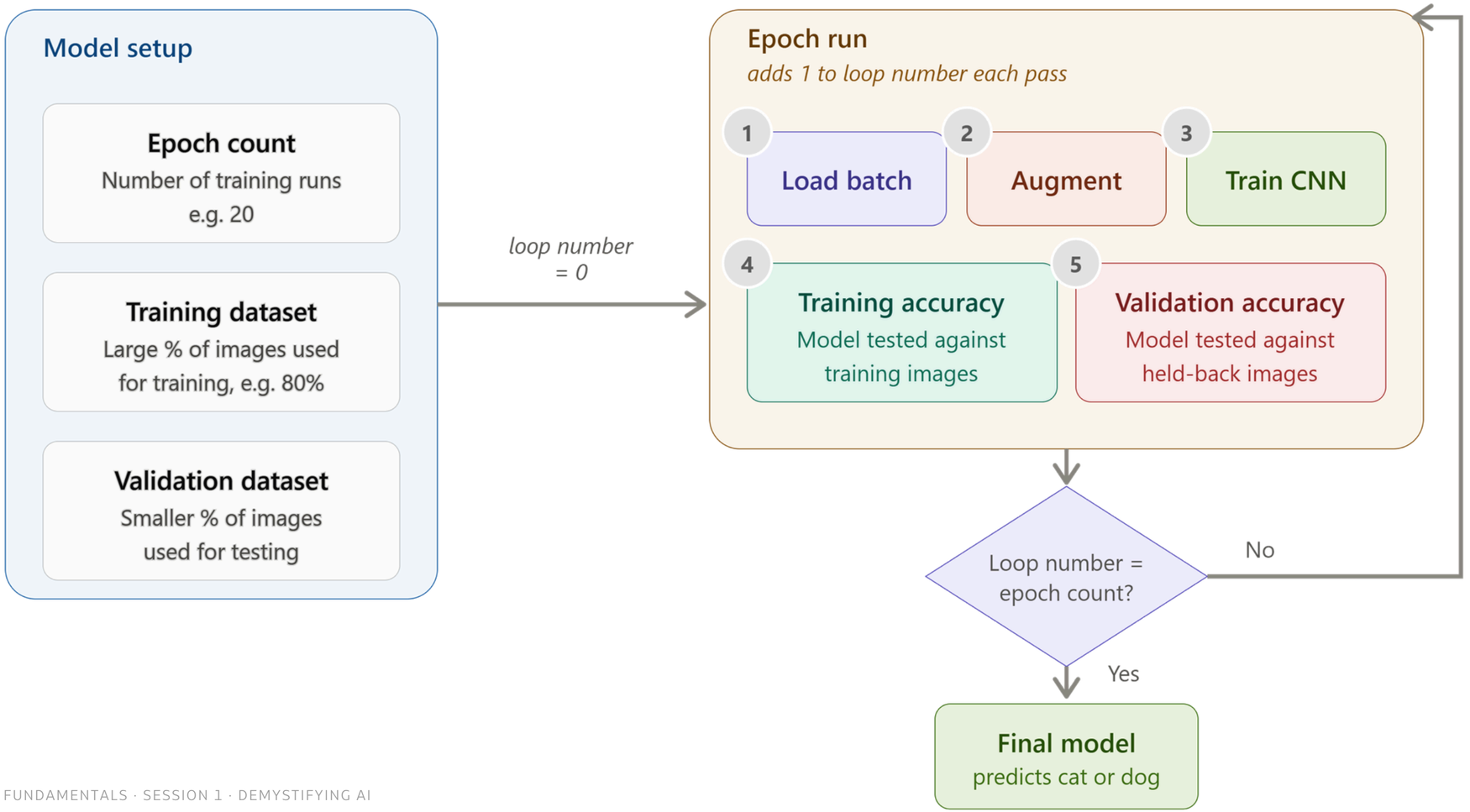
CLASSIFICATION AI

HOW TRAINING WORKS



CLASSIFICATION AI

HOW TRAINING WORKS



CLASSIFICATION AI

EXERCISE CHECKLIST

1. Look through the Model Architecture and Dataset
2. Train the Model
    - Discuss model accuracy metrics
3. Ask the AI to classify this image
4. Feedback/Retraining Loop
    a. Does it get the answer correct? If not, why?
    b. What can we change?
    c. Did the change work?



15-mins Demo

CLASSIFICATION AI

WHAT WE LEARNT

- CNNs can be customised:
  - For example, if we wanted fur colour to be less of an issue we could have reduced the impact by added a layer to further augment data to use random colours and brightness

CODE WINDOW

```
# Colour augmentations

layers.RandomBrightness(0.2),7
layers.RandomContrast(0.2),89
```

- **Importantly:**
  - AI is only as good as its training data
  - It is vital for datasets to be varied and large
    - Especially to allow for a larger number of randomised training images

CLASSIFICATION AI

CONCLUSION

FINAL QUESTION

What two things do you know about this AI?

- Used Supervised Learning to Train
- Example of Traditional AI

This demo showed how:

- Models are created
- Trained
- Improved by retraining

💡 *N.B. Most of you will never need to do this, as you'll typically use pre-trained commercial models.*

DEMYSTIFYING AI

ASSIGNMENT

Coderbyte MCQs
10 questions to evaluate understanding

SESSION OUTCOMES

OBJECTIVES ACHIEVED

- Confidently define AI and its subfields
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NEXT TIME

WHAT TO EXPECT

- Modern AI and how its architecture differs from the traditional AI models
- How to evaluate and optimise AI performance
- Reliability & Limitations

THANK YOU

